10 A student places a sample of an isotope of protactinium (Pa-234) near a radiation detector. The readings on the detector, taken every 20 s, are recorded in Table 10.1.

Table 10.1

time / s	count rate / counts / min
0	101
20	88
40	76
60	66
80	58
100	51
120	46
140	42
160	38
180	35

Fig. 10.1 shows a graph of the count rate **due to this sample** against time.

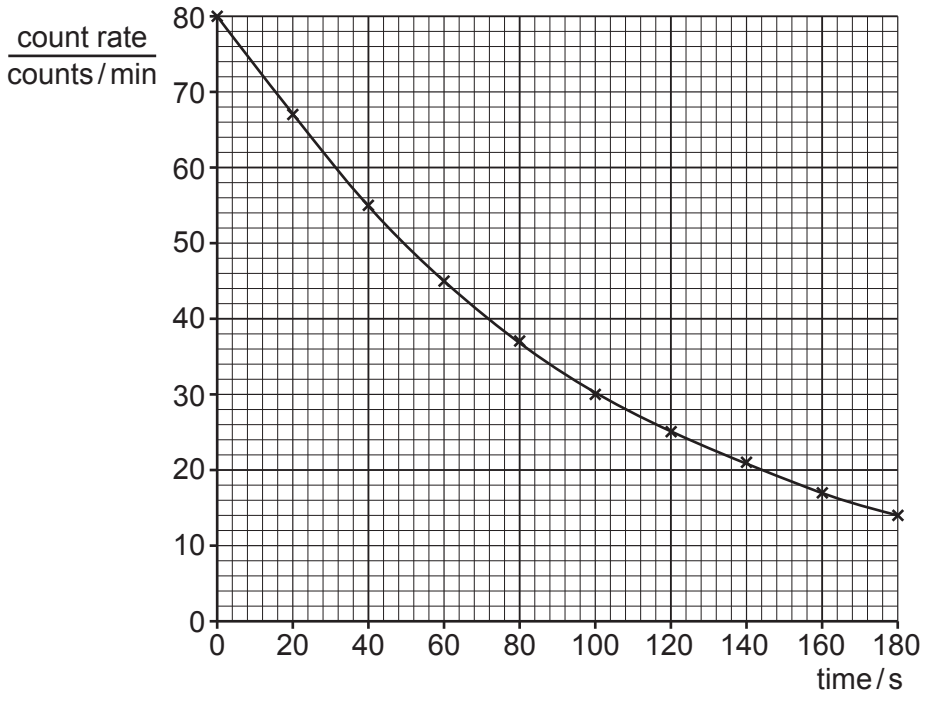


Fig. 10.1

(a) Explain why the readings in Table 10.1 are **not** the same as those plotted on the graph.

.....

..... [2]

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(b) Using the graph in Fig. 10.1, determine the half-life of this isotope of protactinium.

half-life = ... s [2]

(c) The nuclide notation for this isotope of protactinium is $^{234}_{91}Pa$.

Protactinium-234 decays to an isotope of uranium (U) by β-emission.

Write down the nuclide equation for this decay of protactinium-234.

[3]

[Total: 7]

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